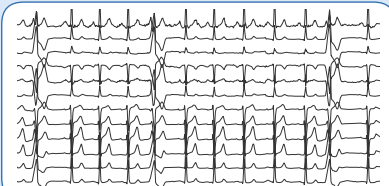
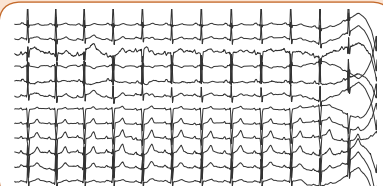


A. Clinical ECG sources

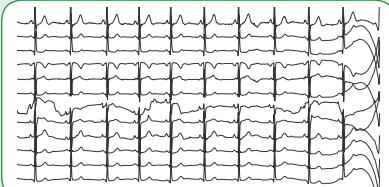
PTB-XL



Georgia



Chapman



Ningbo



Harmonised label space

13 diagnostic labels retained
across all sources | 66,379 recordings

Leave-one-source-out evaluation

Fold 1

Test:

PTB-XL

Fold 2

Test:

Georgia

Fold 3

Test:

Chapman

Fold 4

Test:

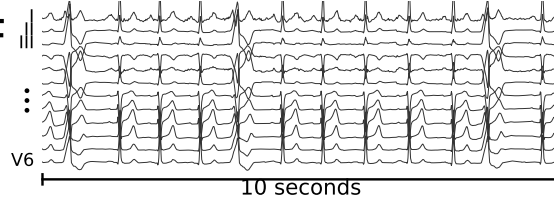
Ningbo

train on three sources, test on the fourth
(four rotations)

B. Lightweight architecture benchmark

12-lead ECG input

10 s at 100 Hz
(12 x 1,000 samples)



10 encoder families × 2 size budgets

FCN-1D



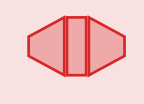
ResNet-1D



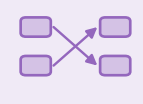
TCN



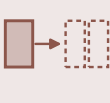
MobileNetV2



ShuffleNetV2



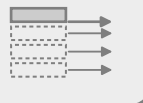
GhostNet



MobileNetV3



FasterNet



RepViT



MobileNetV4 (Conv)



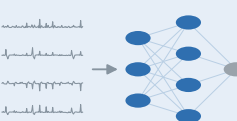
Compact

≤ 32 KB INT8 weights

Standard

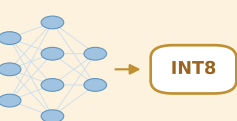
≤ 128 KB INT8 weights

Train (FP32)



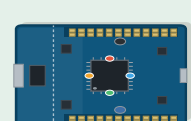
same protocol for
all models and sources

Quantise (INT8)



post-training
quantisation

Deploy (STM32F411)

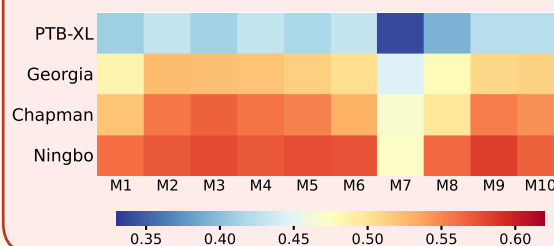


measured on device:
Flash, SRAM, latency

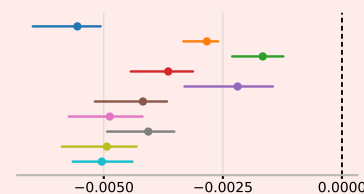
C. Evaluation outputs

RQ1. External discrimination

macro-AUPRC (FP32)

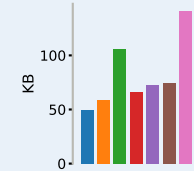


INT8 - FP32 (equal-source)

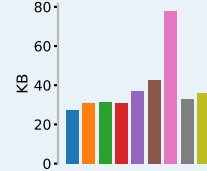


RQ2. Hardware characterisation

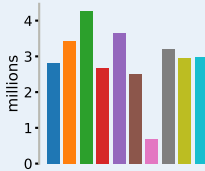
Flash



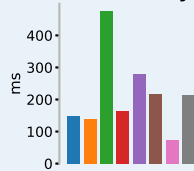
Peak SRAM



MACCs

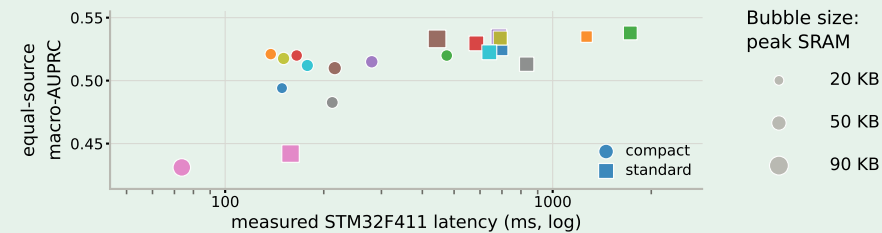


Latency



- measured Flash, peak SRAM, MACCs and latency on STM32F411
- both size budgets for each architecture family

RQ3. Performance-resource trade-offs



- external macro-AUPRC against measured latency
- SRAM shown as bubble size; both budgets drawn